

An Introduction to the Theory of Programming Languages

LAMA (post)doc days

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1 Introduction

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Remark: “Skeletal” is for next time 😬, “Mathematical” (for my PhD) is category theory

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 - ✕ more powerful !

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 - do some cool math 🧐

2 λ -calculus

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Claim (Church-Turing Thesis): A mathematical function on the natural numbers can be calculated by an *effective* method if and only if it can be defined in the λ -calculus (if and only if it can be computed by a Turing machine).

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The rest of this talk:

1. define λ -calculus
2. show some “simple” results
3. introduce a typed version of λ -calculus
4. present the strong normalization of this new calculus and why it’s nice

2.2 Syntax

Definition (λ -terms):

$$t, u \dots ::= x \mid t \ u \mid \lambda x. t$$

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Definition (Normal Form): t is a **normal form** if there is no u such that $t \rightarrow_{\beta} u$. We write $t \nrightarrow_{\beta}$.

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Definition: $t \rightarrow_{\beta}^* u \Leftrightarrow t \rightarrow_{\beta} \dots \rightarrow_{\beta} u$

Theorem (Confluence): For any terms t, u, u' such that $t \rightarrow_{\beta} u$ and $t \rightarrow_{\beta} u'$, there exists a term v such that $u \rightarrow_{\beta}^* v$ and $u' \rightarrow_{\beta}^* v$.

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Corollary: If $t \rightarrow_{\beta}^* u \nrightarrow_{\beta}$ then such a u is unique.

Proof of Confluence: Introduce a new relation $t \twoheadrightarrow_{\beta} u$, show it has the “diamond property” and show it entails confluence for $\rightarrow_{\beta}$. See rest of this section. $\square$

2.4 Confluence

Definition (Simultaneous reduction):

$$\frac{t \twoheadrightarrow_{\beta} t' \quad u \twoheadrightarrow_{\beta} u'}{(\lambda x.t) u \twoheadrightarrow_{\beta} t'[x/u']} \beta \quad \frac{t \twoheadrightarrow_{\beta} t'}{\lambda x.t \twoheadrightarrow_{\beta} \lambda x.t'} \lambda \quad \frac{t \twoheadrightarrow_{\beta} t' \quad u \twoheadrightarrow_{\beta} u'}{t u \twoheadrightarrow_{\beta} t' u'} \text{app} \quad \frac{}{t \twoheadrightarrow_{\beta} t} \text{id}$$

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Lemma: $t \rightarrow_{\beta} t' \implies t \twoheadrightarrow_{\beta} t' \implies t \rightarrow_{\beta}^* t'.$

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Lemma: $t \rightarrow_{\beta} t' \implies t \twoheadrightarrow_{\beta} t' \implies t \rightarrow_{\beta}^* t'.$

Lemma (Diamond property): For any terms t, u, u' such that $t \twoheadrightarrow_{\beta} u$ and $t \twoheadrightarrow_{\beta} u'$, there exists a term v such that $u \twoheadrightarrow_{\beta} v$ and $u' \twoheadrightarrow_{\beta} v$.

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Proof of confluence:

$$\begin{aligned} t \rightarrow_{\beta} u \wedge t \rightarrow_{\beta} u' &\implies t \twoheadrightarrow_{\beta} u \wedge t \twoheadrightarrow_{\beta} u' \\ &\implies \exists v, u \twoheadrightarrow_{\beta} v \wedge u' \twoheadrightarrow_{\beta} v \\ &\implies \exists v, u \rightarrow_{\beta}^* v \wedge u' \rightarrow_{\beta}^* v \end{aligned}$$



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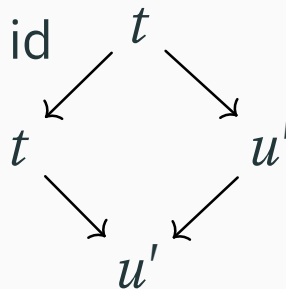
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□

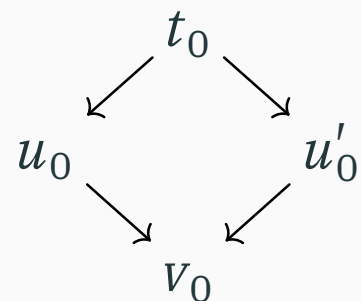
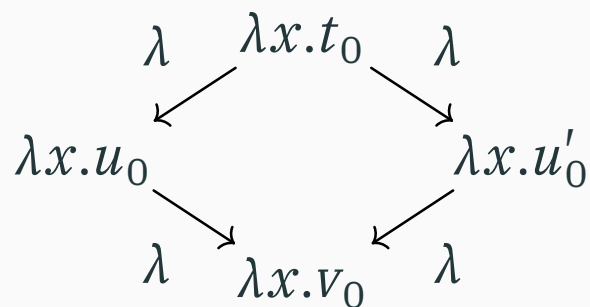
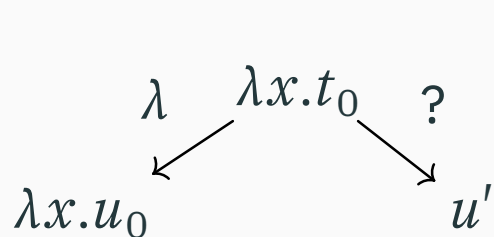
Proof of the Diamond Property lemma: By induction on $t \twoheadrightarrow_{\beta} u$.

id vs. anything $u = t$



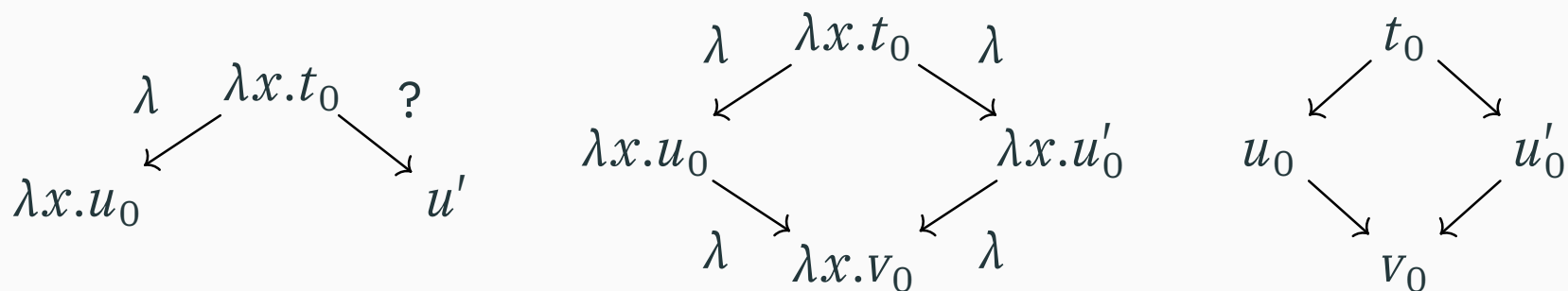
2.4 Confluence

λ vs. anything $t = \lambda x.t_0$, $u = \lambda x.u_0$ and $t_0 \rightarrow_{\beta} u_0$

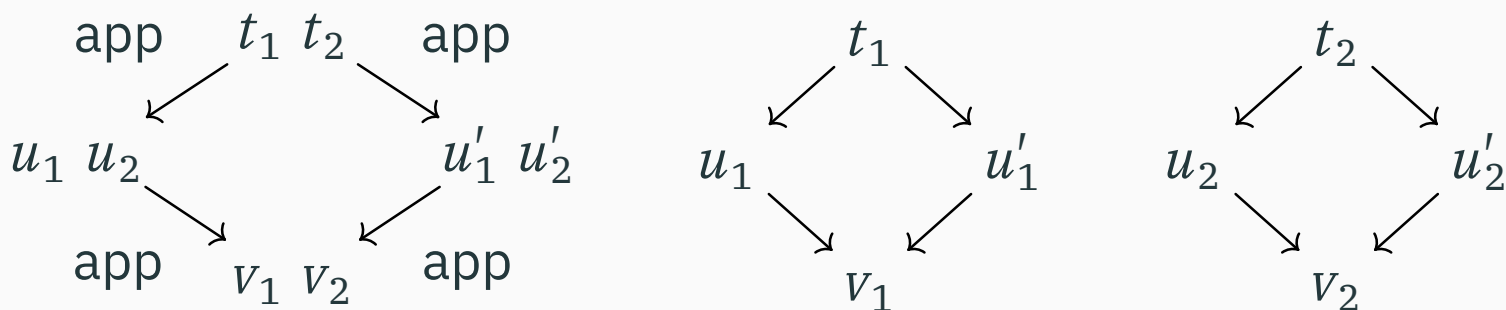


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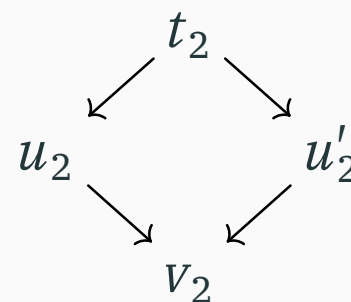
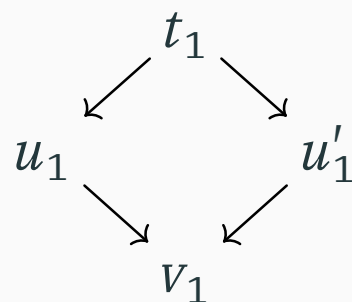
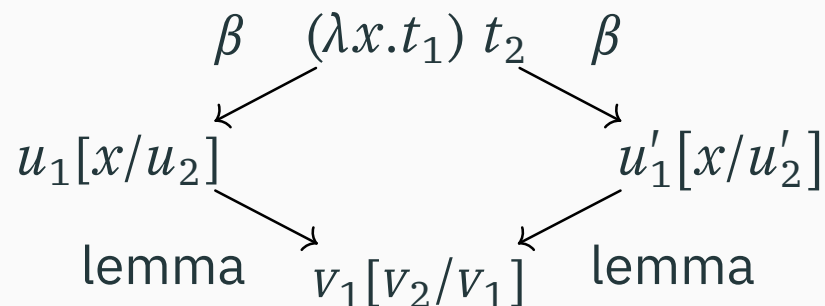


app vs. app $t = t_1 t_2$, $u = u_1 u_2$, $u' = u'_1 u'_2$ and ...



2.4 Confluence

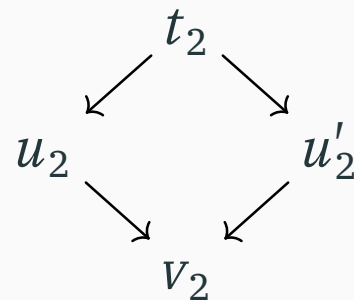
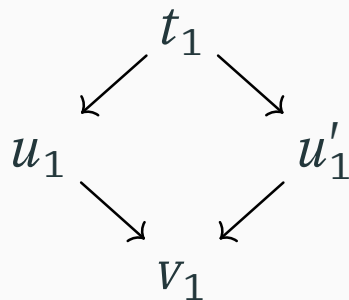
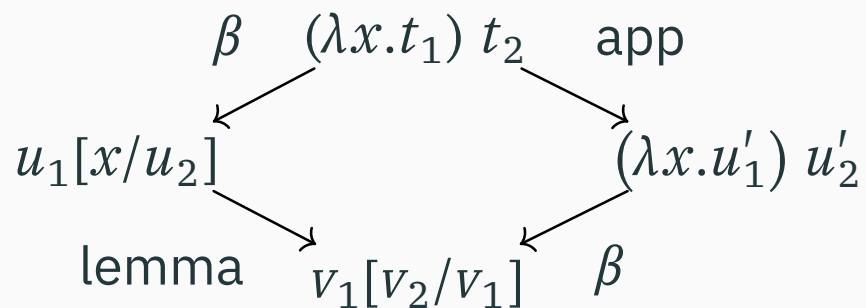
β vs. β $t = (\lambda x.t_1) t_2$ and ...



Lemma: $t \twoheadrightarrow_{\beta} t' \wedge u \twoheadrightarrow_{\beta} u' \Rightarrow t[x/u] \twoheadrightarrow_{\beta} t'[x/u']$.

2.4 Confluence

β vs. app $t = (\lambda x.t_1) t_2$ and ...



3 Simply Typed λ -calculus

3.1 Types

Idea:

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unit bool nat $\rightarrow$ nat bool $\rightarrow$ unit nat $\rightarrow$ (nat $\rightarrow$ nat)

(nat $\rightarrow$ bool) $\rightarrow$ (bool $\rightarrow$ nat) unit $\rightarrow$ ((bool $\rightarrow$ unit) $\rightarrow$ nat)

3.2 Typing

Definition (Context): List of “typed variables” $(x : A) :$

$$\Gamma ::= (x : A), (y : B), (z : C) \dots$$

Definition (Typing judgment): “Term t has type A in context Γ ”

$$\Gamma \vdash t : A$$

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Definition (Typing rules):

$$\frac{(x : A) \in \Gamma}{\Gamma \vdash x : A} \text{var}$$

$$\frac{\Gamma \vdash t : A \rightarrow B \quad \Gamma \vdash u : A}{\Gamma \vdash t u : B} \text{app}$$

$$\frac{\Gamma, (x : A) \vdash t : B}{\Gamma \vdash \lambda x. t : A \rightarrow B} \lambda$$

3.2 Typing

Example: $\emptyset \vdash \lambda x.x : ?$

$\emptyset \vdash \lambda x.\lambda y.y\ x : ?$

$k : ?, z : ? \vdash (\lambda x.\lambda y.z\ x)\ k : ?$

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$$\frac{\frac{\frac{(x : A), (y : A \rightarrow B) \vdash y : A \rightarrow B \quad (x : A), (y : A \rightarrow B) \vdash x : A}{\text{app}}}{(x : A), (y : A \rightarrow B) \vdash y \ x : B}{\lambda} \quad \lambda$$

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$$\emptyset \vdash \lambda x. \lambda y. y \ x : A \rightarrow (A \rightarrow B) \rightarrow B$$
$$\begin{array}{c}
\text{var} \quad \frac{}{..., z : A \rightarrow B \vdash z : A \rightarrow B} \quad \text{var} \quad \frac{}{..., x : A \vdash x : A} \\
\text{app} \quad \frac{}{} \\
\lambda \quad \frac{k : A, z : A \rightarrow B, x : A, y : C \vdash z x : B}{k : A, z : A \rightarrow B, x : A \vdash \lambda y. z x : C \rightarrow B} \\
\lambda \quad \frac{k : A, z : A \rightarrow B, x : A \vdash \lambda y. z x : C \rightarrow B}{k : A, z : A \rightarrow B \vdash \lambda x. \lambda y. z x : A \rightarrow (C \rightarrow B)} \quad \text{var} \quad \frac{}{k : A, z : A \rightarrow B \vdash k : A} \\
\text{app} \quad \frac{k : A, z : A \rightarrow B \vdash \lambda x. \lambda y. z x : A \rightarrow (C \rightarrow B) \quad k : A, z : A \rightarrow B \vdash k : A}{k : A, z : A \rightarrow B \vdash (\lambda x. \lambda y. z x) k : C \rightarrow B}
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Remark: Recall $\Omega = \omega \ \omega$ with $\omega = \lambda x.x \ x$ such that $\Omega \rightarrow_{\beta} \Omega \rightarrow_{\beta} \dots$. What is the type of ω ?

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$C \rightarrow B = A = C \rightsquigarrow$ impossible, ω cannot be typed ! Therefore, $\Omega = \omega \ \omega$ can't be typed either.

3.3 Strong Normalization

Theorem: Simply typed λ -calculus is strongly normalizing:

$$\forall t, (\exists \Gamma, \exists A, \Gamma \vdash t : A) \implies (\exists u, t \rightarrow_{\beta}^* u \nrightarrow_{\beta})$$

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